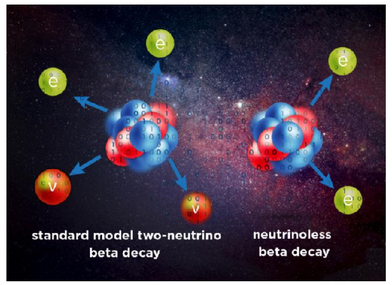
Gravitational Wave Paleontology

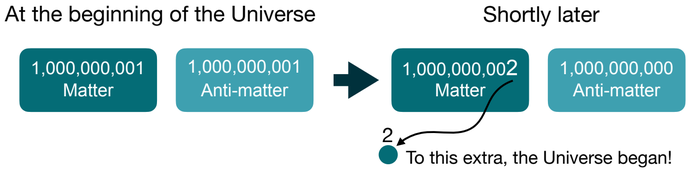
Digging in Spacetime with AI & Simulation

Ann-Kathrin Schütz
Lawrence Berkeley National Laboratory

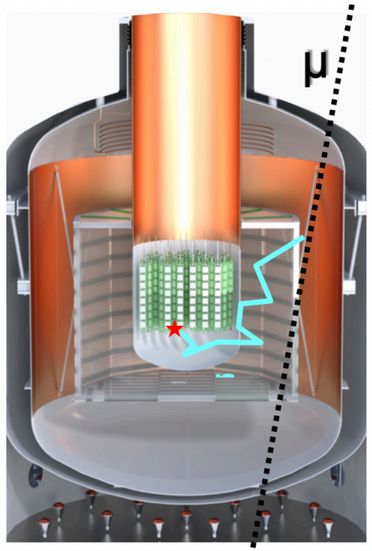
Motivation - Physics Design Optimization

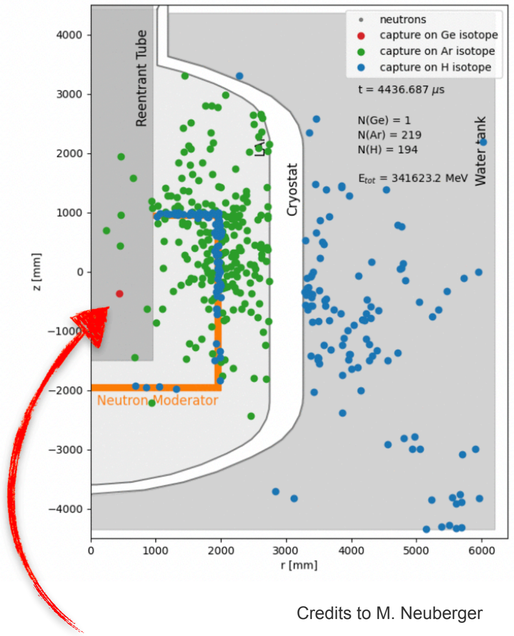


Unlocking the Secrets of the Universe through Neutrinoless Double Beta Decay



- Must suppress **extremely rare backgrounds** from cosmic muon-induced neutrons.
- Simulations are **expensive**, signal is **vanishingly rare.**
- Optimizing designs under such rare-event statistics is intractable.
- Need a rare event surrogate model for efficient, variance-aware optimization.





Credits to M. Neuberger

very rare dangerous event

Rare Event Surrogate Model (RESuM) for Physics Detector Design

- **66.5% reduction in neutron background** with **uncertainty quantification**
- **only 3.3% of the computational resources** compared to traditional method.

ICLR International Conference On Learning Representations
Spotlight

Alan Poon Aobo Li

Finding a needle in a haystack...

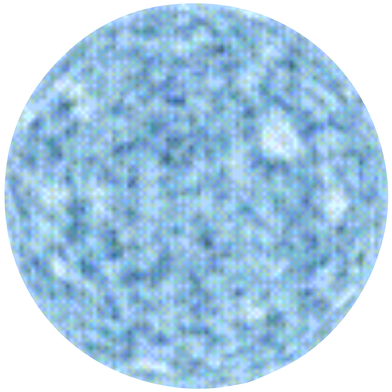
"Rare Events are everywhere"

Examples:

- Direct detection of WIMP dark matter
  <O(1) events/(30 kg years)
- Neutrinoless double beta decay (0vbb)
  <0(1) events/(500 kg year)
- Neutrino coherent scattering (CEvNS)
  ~0(1) events/(10 kg years)
- Binary Black Hole Mergers

"Rare Events are everywhere"

It is challenging to observe Massive Stars ...



Massive Stars are **Rare**

Number of stars $\propto$ Mass$^{-2.3}$

(Salpeter 1955)

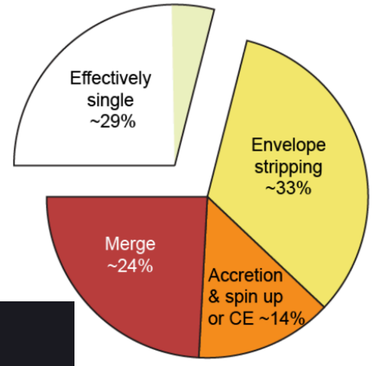
Massive stars live fast and die young

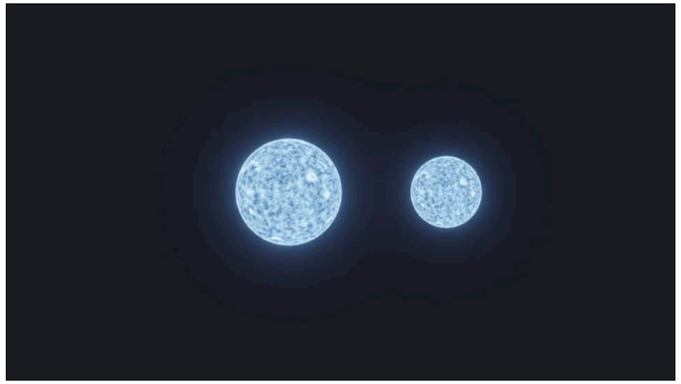
Lifetime $\propto$ Mass$^{-2.8}$

(Kippenhanh & Weigert 1990)

They live in binaries

~70% of massive stars will interact with a companion during their lifetime (Sana + 2012)





Sana + (2012), Moe & DiStefano + (2017)

Binary interactions are crucial in forming close black hole binaries which can merge and create gravitational waves.

Isolated binaries

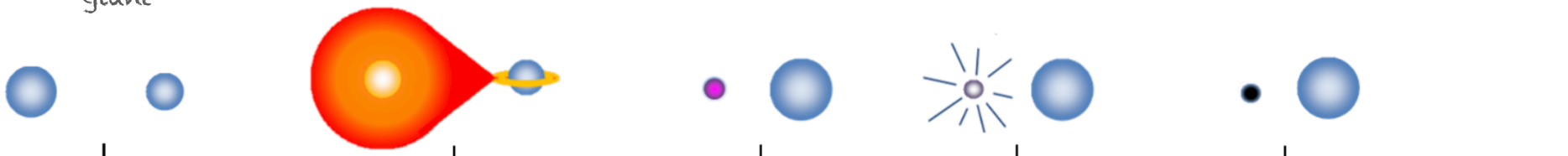
More massive star 1 evokes faster, grows in size and becomes a red giant

Material flows from red giant onto its companion

Star 1 sheds its envelope, leaving behind a helium star

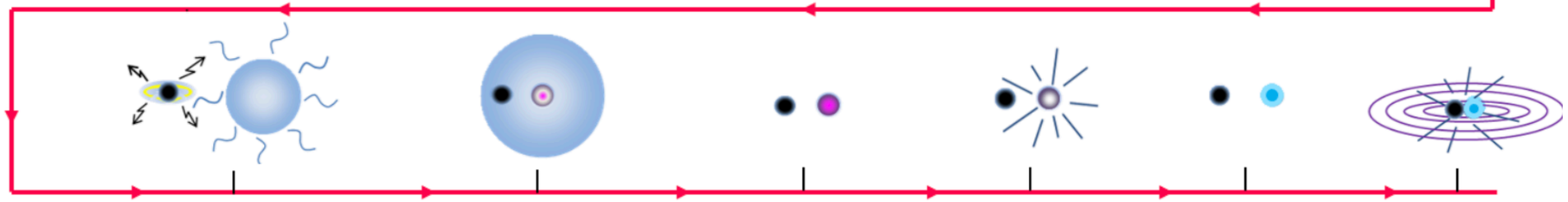
Star 1 goes SN

First compact object formed



Zero age | Mass Transfer | Supernova 1 | BH

Second star evolves,



X-ray binary | Common Envelope | Supernova 2 | BH-NS

X-ray binary, with BH accreting material, star 2 expands

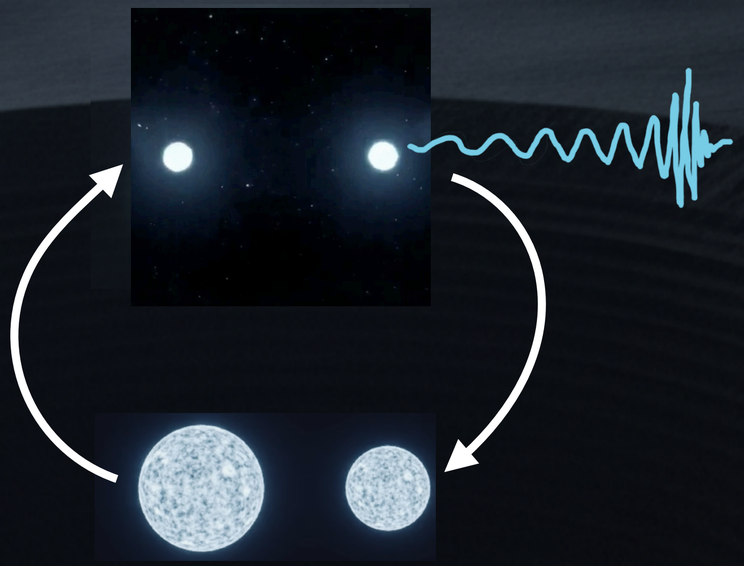
Unstable and rapid mass transfer, BH enters envelope, Drag forces → BH spirals inward, Energy is transferred to envelop

Envelop ejected, leaving BH & stripped star with much tighter orbit

Second star goes SN, forming DCO

DCOs can eventually spiral in and merge forming GW

What we see

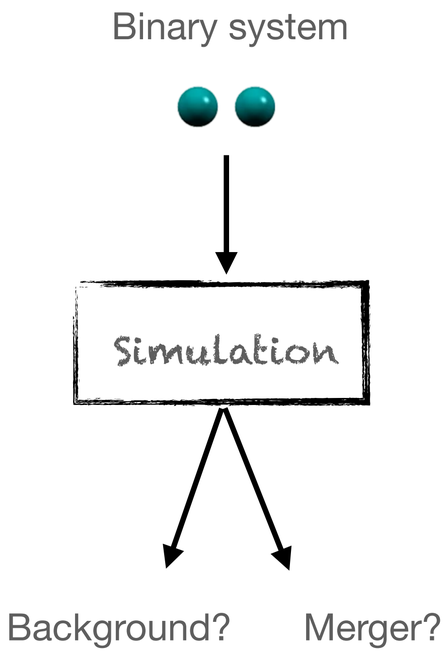


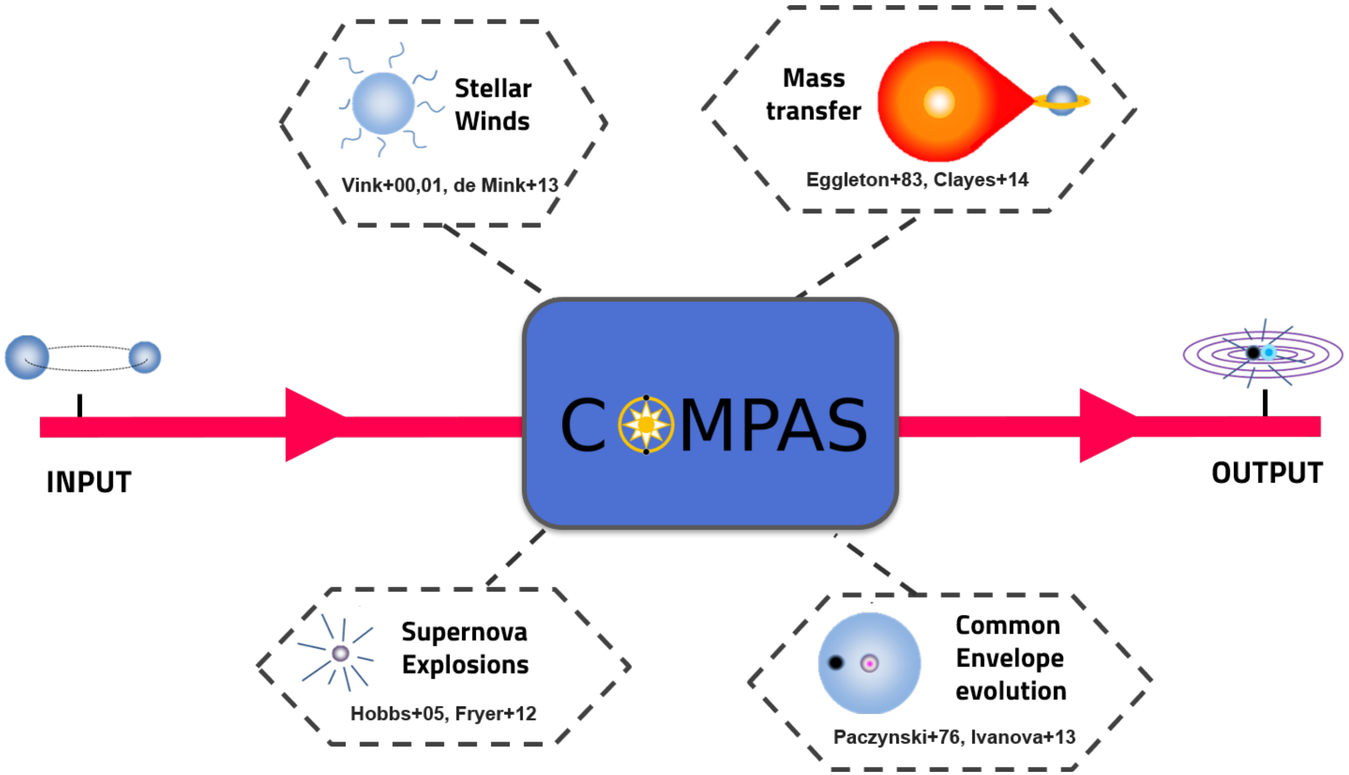
Stars in the Past

Gravitational Wave Paleontology

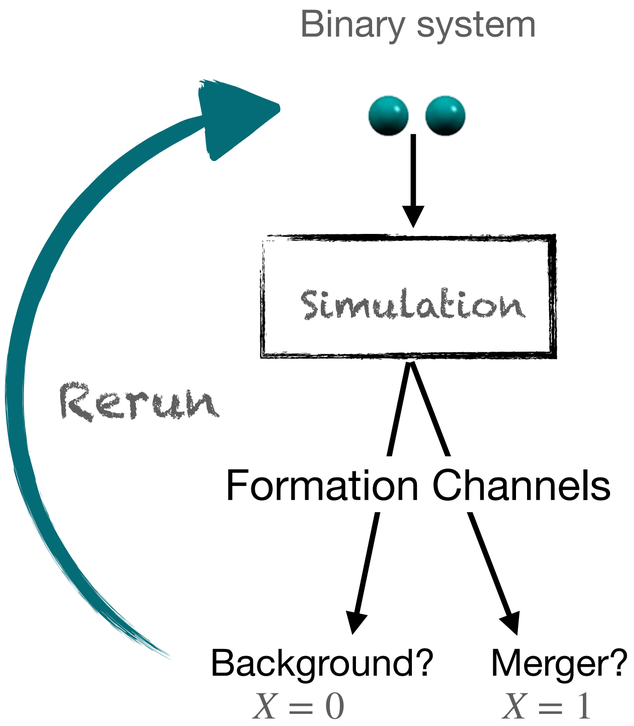
... like cosmic forensics

Prof.
Floor Broekgaarden
UCSD
et. al.

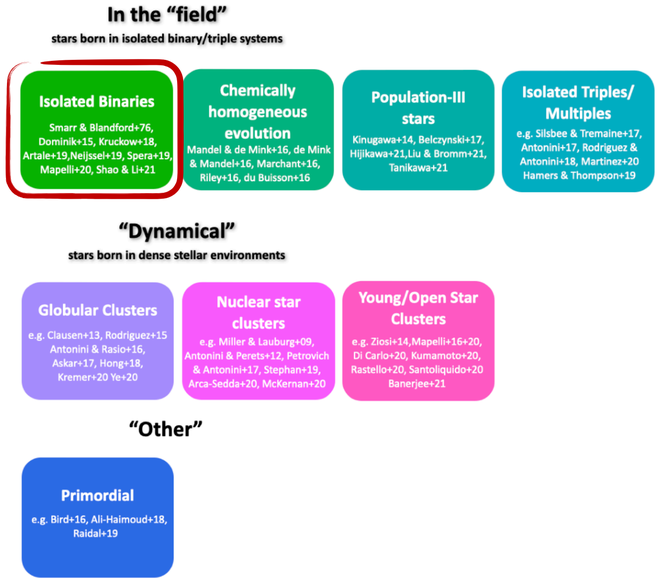

Binary system
Simulation
Background?
Merger?

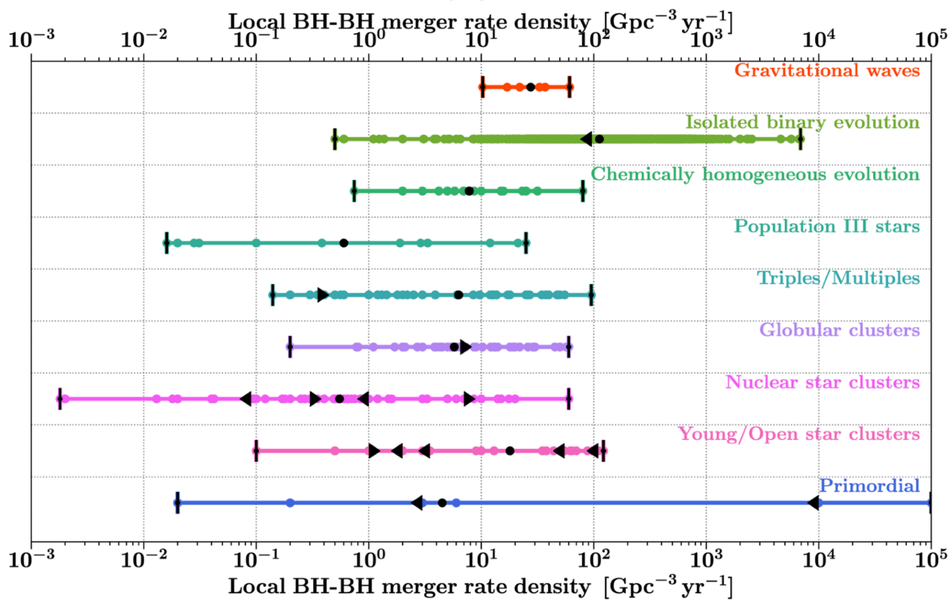

Stellar Winds
Vink+00,01, de Mink+13
Mass transfer
Eggleton+83, Claves+14
INPUT
COMPAS
OUTPUT
Supernova Explosions
Hobbs+05, Fryer+12
Common Envelope evolution
Paczynski+76, Ivanova+13

This talk **only isolated Binaries** - Inclusion of additional formation channels → future work



Merger rate?



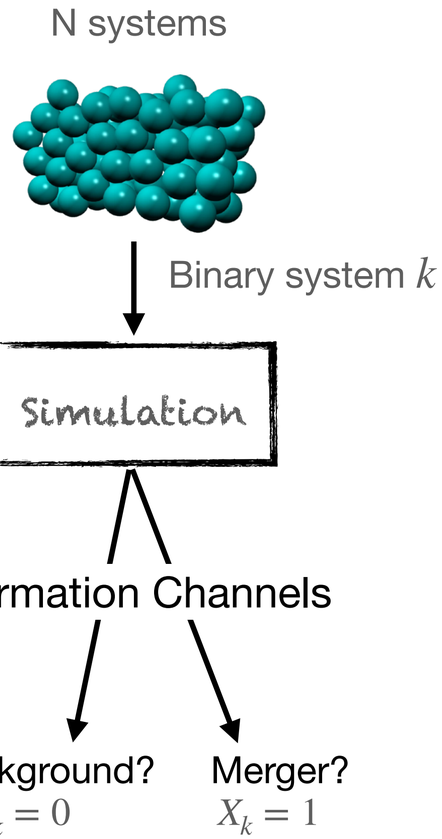


[Broekgaarden et al. (2021)]

Many different formation channels!

!! Rerun simulations for many different formation channels !!

Initial Conditions



Merger rate?

$$\epsilon = \frac{1}{N} \sum X_k$$

Binary fraction

IMF

Initial mass ratios

Initial Periods

Initial orbital separation?

Birth Metallicities

Initial eccentricity?

Properties Environment

Initial Rotation?

Initial conditions: x

m_1: the mass of the initially more massive star	$[8, 150]\ M_\odot$	Power law(-2.35)
m_2: the mass of the initially less massive star	$(0.1\ M_\odot, m_1]$	Uniform
a: the initial orbital separation	[0.01, 1000] AU	Power law(-1)
$\mathbf{v}_i$: supernova natal kick vector for supernova i, $i = 1, 2$, including:		
v_i—magnitude of the supernova natal kick ($\mathrm{km\,s^{-1}}$)	$[0, \infty)$	Maxwellian
θ_i—polar angle defining the direction of the natal kick	$[0, \pi]$	Uniform
ϕ_i—azimuthal angle defining the direction of the natal kick	$[0, 2\pi]$	Uniform
ω_i—mean anomaly	$[0, 2\pi]$	Uniform

Each system has system-specific parameters $\phi_{k,i}$ (~ 30 parameters)

!! Rerun simulations for many different initial conditions!!

Rare Event Problem

N binary systems with

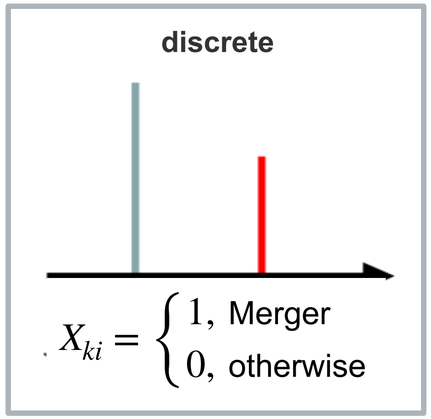
- hyper-parameters $\boldsymbol{\theta}_k$ and
- initial condition parameters $\boldsymbol{\phi}_{ik}$

Fixed in a simulation run of N binary systems, global

binary system specific, varies for each system, local

System Outcome

X_{ik}



$$X_{ik} \sim \text{Bernoulli}(p = t(\boldsymbol{\theta}_k, \boldsymbol{\phi}_{ik}))$$

Merger Formation Probability

$t(\boldsymbol{\theta}_k, \boldsymbol{\phi}_{i,k})$ **small**

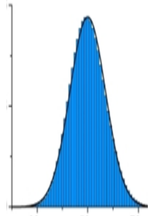
Merger rate

$$y = \frac{1}{N} \sum X_{ik}$$

$$\sim \text{Poisson}(N\bar{t})/N$$

$$y \ll 1$$

Large N



Large N scenario

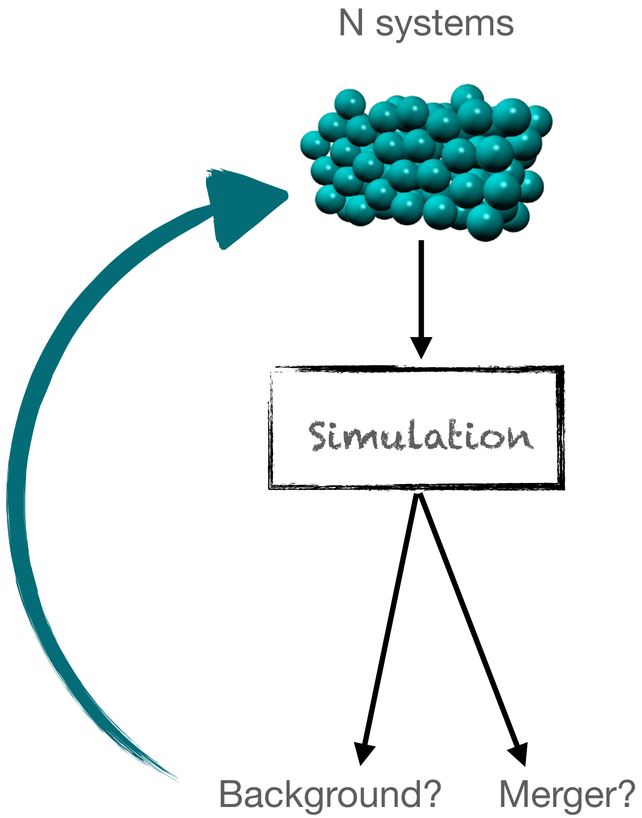
$$y \sim \mathcal{N}(\bar{t}(\boldsymbol{\theta}), \frac{\bar{t}(\boldsymbol{\theta})}{N})$$

Low Variance, High Cost.

$N \to \infty$, y will asymptotically approximate the **expected formation probability**

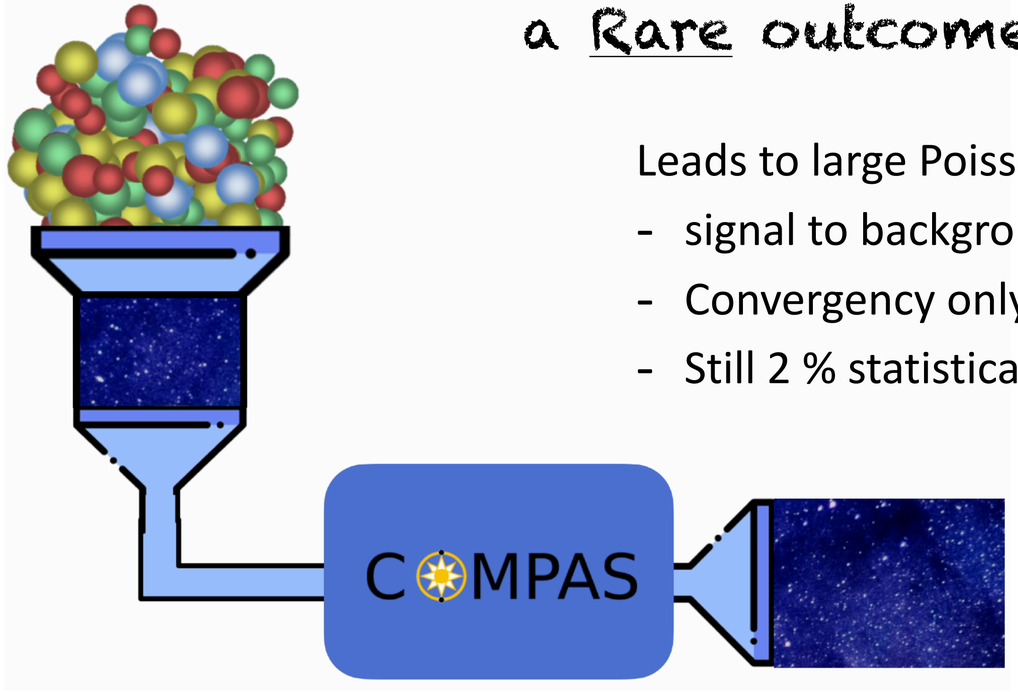
$$\bar{t}(\boldsymbol{\theta}) = \int t(\boldsymbol{\theta}, \boldsymbol{\phi}) g(\boldsymbol{\phi}) \mathrm{d}\boldsymbol{\phi}$$

Sampling Uncertainty



Merger rate?

1 000 000 binaries



Gravitational-wave sources are a Rare outcome among stars

Leads to large Poisson (sampling) noise

- signal to background ratio $1:10^6$
- Convergency only for $N > 3 \cdot 10^8$
- Still 2 % statistical noise

only ~1 merger!

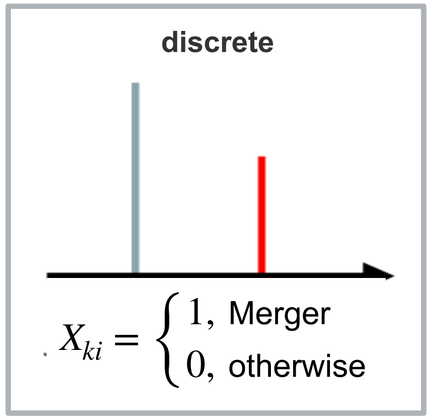
!! Rerun simulations to reduce statistical noise !!

N binary systems with

- parameters of interest $\boldsymbol{\theta}_k$ and
- Initial condition parameters $\boldsymbol{\phi}_{ik}$

System Outcome

X_{ik}



$$X_{ik} \sim \text{Bernoulli}(p = t(\boldsymbol{\theta}_k, \boldsymbol{\phi}_{ik}))$$

Merger Formation Probability
$t(\boldsymbol{\theta}_k, \boldsymbol{\phi}_{i,k})$ **small**

Merger rate

$$y = \frac{1}{N} \sum X_{ik}$$

$$\sim \text{Poisson}(N\bar{t})/N$$

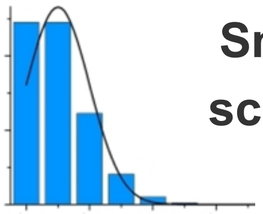
$$y \ll 1$$

Small N

Expensive simulator

Small N Scenario as simulation is costly:

- y very sensitive to statistical fluctuations.
- y takes **discrete values** and cannot be approximated by a normal distribution.



Small N scenario

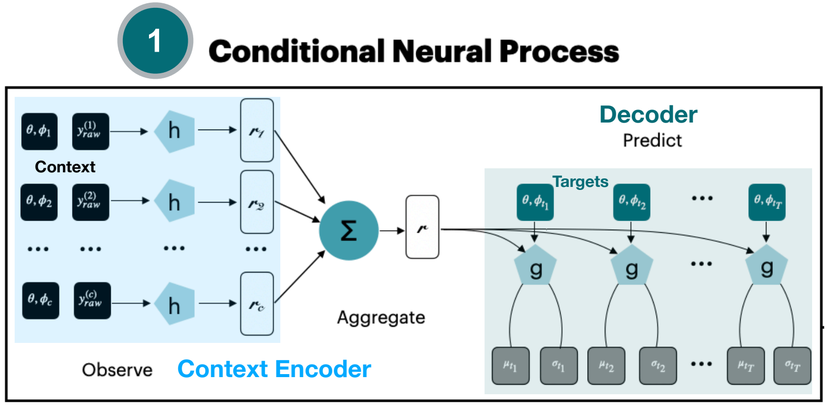
$y \in \{0/N, 1/N, \ldots\}$ discrete

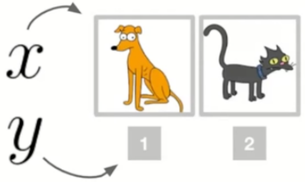
High Variance, **Low Cost**.

Prof.
Floor Broekgaarden
UCSD
et. al.

Ann-Kathrin
Schuetz

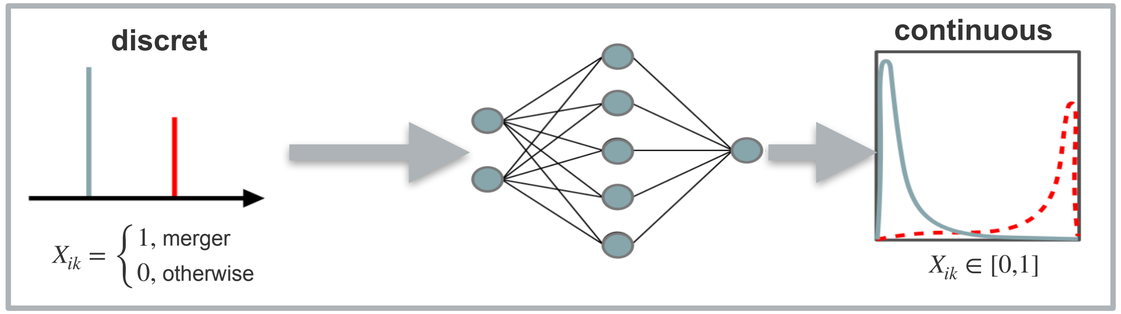
Stars in the Past
COMPAS





- Learn Contextual Features
- Maximize the Posterior Likelihood to Train
- Uncertainty prediction
- Small dataset size (where avoiding overfitting important)

1. Conditional Neutral Process



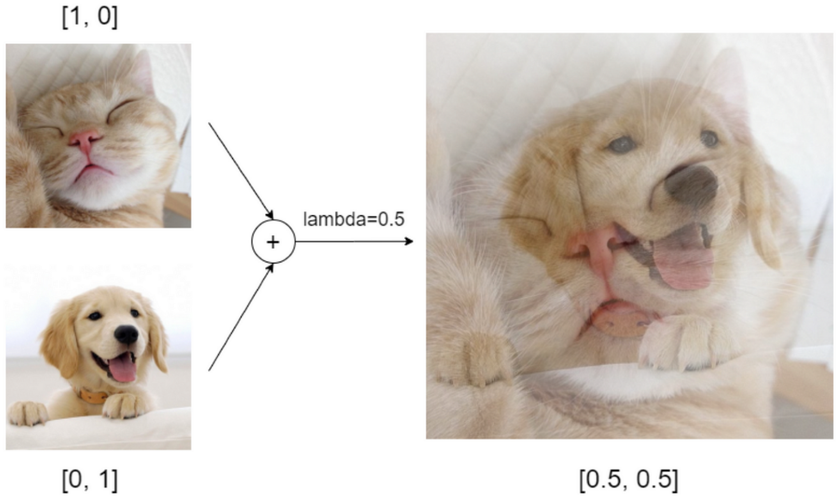
Mitigate statistical noise

- Converts each discrete event outcome into a continuous score
- Propagates uncertainty awareness into final mapping

Problem: severely imbalanced training dataset

Solution: Linear Combinations with Mixup

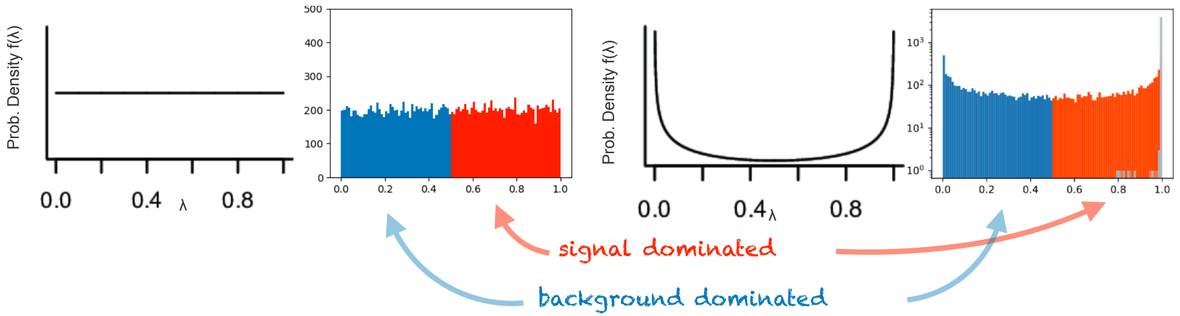
- Create virtual training samples by linearly combining inputs and labels from different, randomly selected classes.
- Linear interpolation in input space should correspond to linear interpolation in output (label) space.
- Smoother decision boundaries and robustness
- Enhances model generalization on test samples that only slightly differ from training data.



$$x = \lambda \cdot x_i + (1 - \lambda) \cdot x_j$$

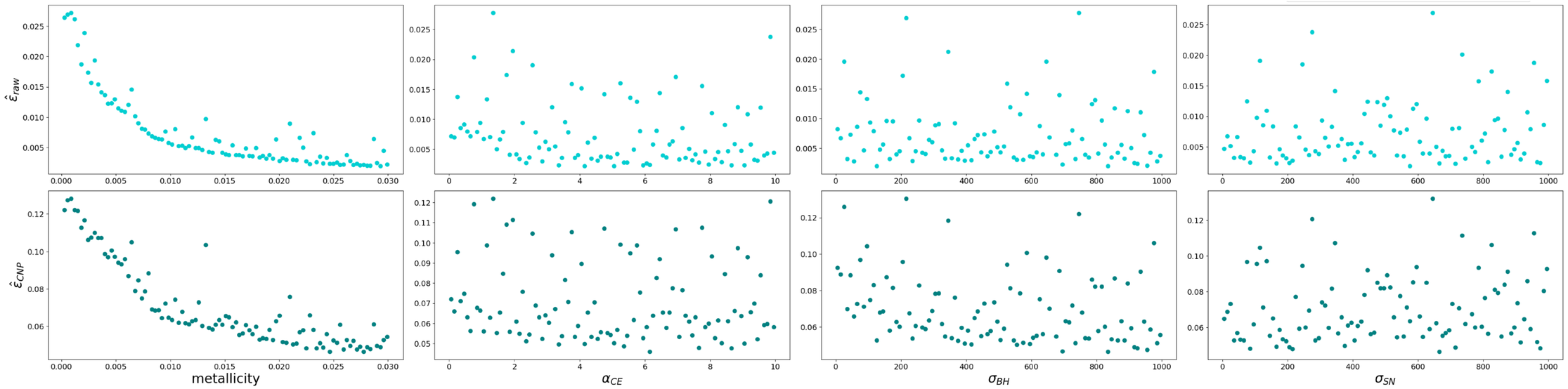
$$y = \lambda \cdot y_i + (1 - \lambda) \cdot y_j$$

Option 1: λ uniformly distr. **Option 2: λ distr. u-shaped**



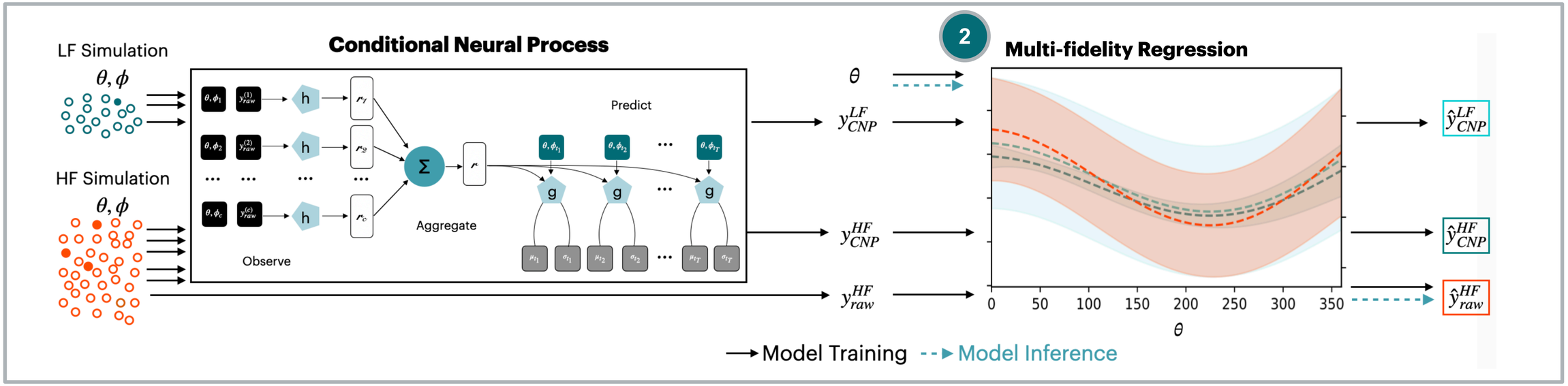
Before

After



- Before CNP: shows significant vertical scatter (noise)
- After CNP: shows clear trend structures with significantly reduced variance.

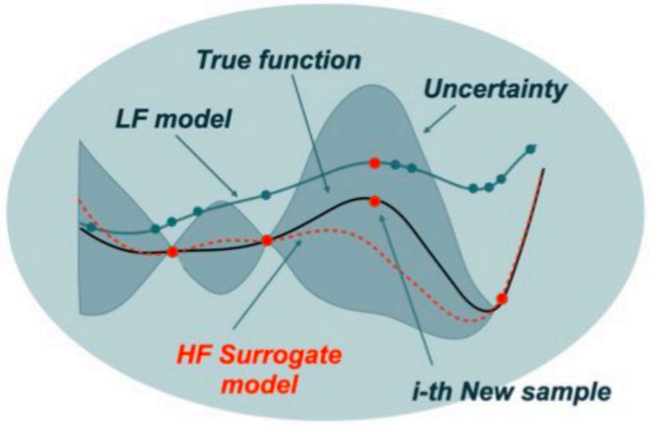
➡ Previously hidden relationships become visible post-smoothing.



Reduce computational cost

- **Multiple simulation fidelities** which different in physical model details and cost
- where low-fidelity (LF) helps with **space exploration**

2. Multi-fidelity Regression



Basic concept

$$y^{(f_i)} = \rho y^{(f_{i-1})} + \delta$$

High Fidelity (HF)

Scaling

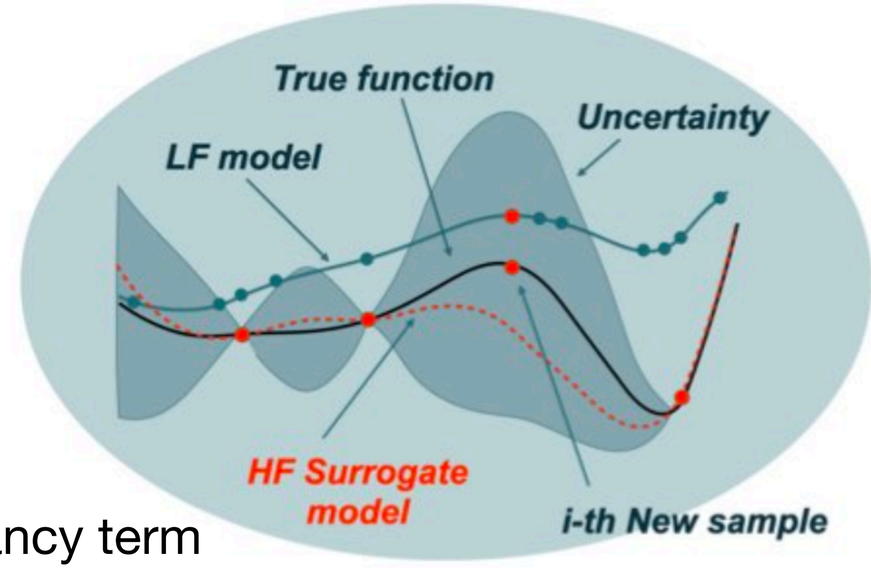
Low Fidelity (LF) modeled by e.g.

Gaussian Process

Polynomial

…

Many possibilities
Smaller N, simplified physical model, broader gird, staged simulation…,
But must share most important underlying principles



Discrepancy term modeled by e.g.

Gaussian Process

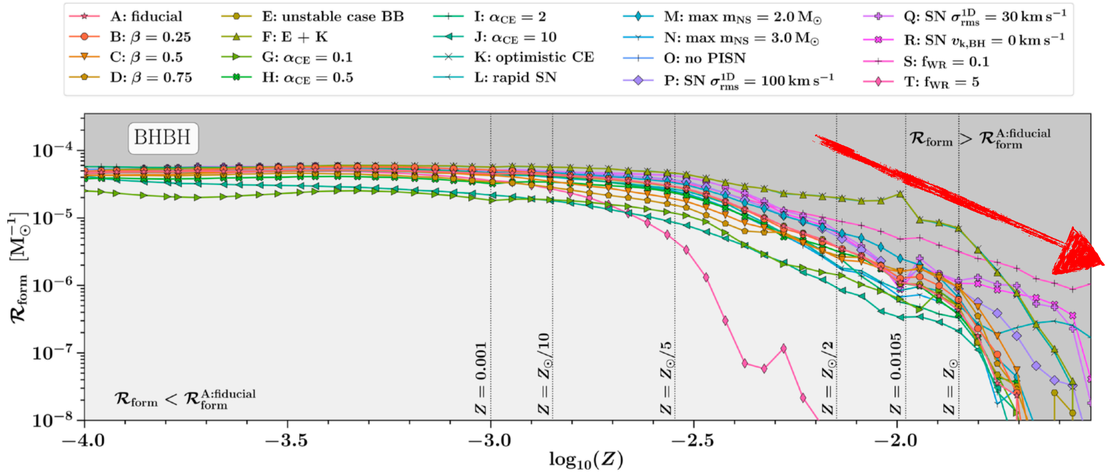
Polynomial

…

We choose

LF: Less initial systems $N^{\mathrm{LF}} < N^{\mathrm{HF}}$ 1000 : 1000000

What do we know about the dependance of the BBH merger rate on metallicity?



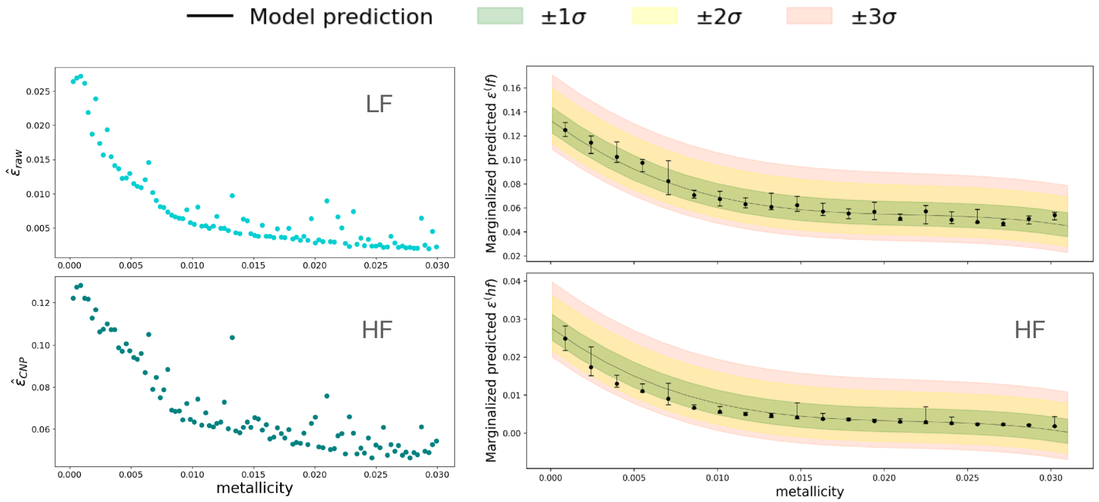
[Broekgaarden et al. (2022)]

- BBH formation rate **decreases strongly with metallicity** (due to weaker stellar winds and tighter binaries)
- The **local universe is more metal-rich** → lower BBH formation rates
- **Early low-metallicity** star formation **dominates present-day BBH** mergers due to delay time

Metallicity strongly affects both **single and binary star evolution** through its impact on **stellar winds**

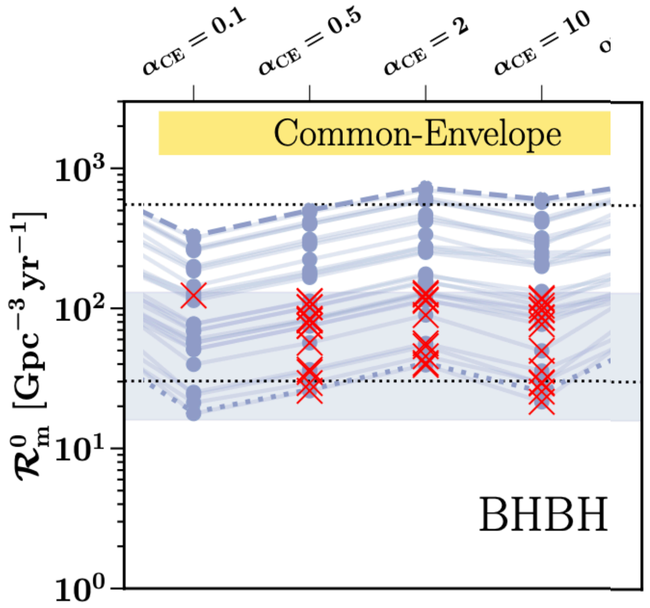
(e.g. higher metallicity → stronger stellar winds → greater mass loss during stellar evolution)

Model Result



- Model confirms strong metallicity dependence
- HF result shows no leveling at high Z, likely due to data sparsity

What do we know about the dependance of the BBH merger rate on the common envelope efficiency?

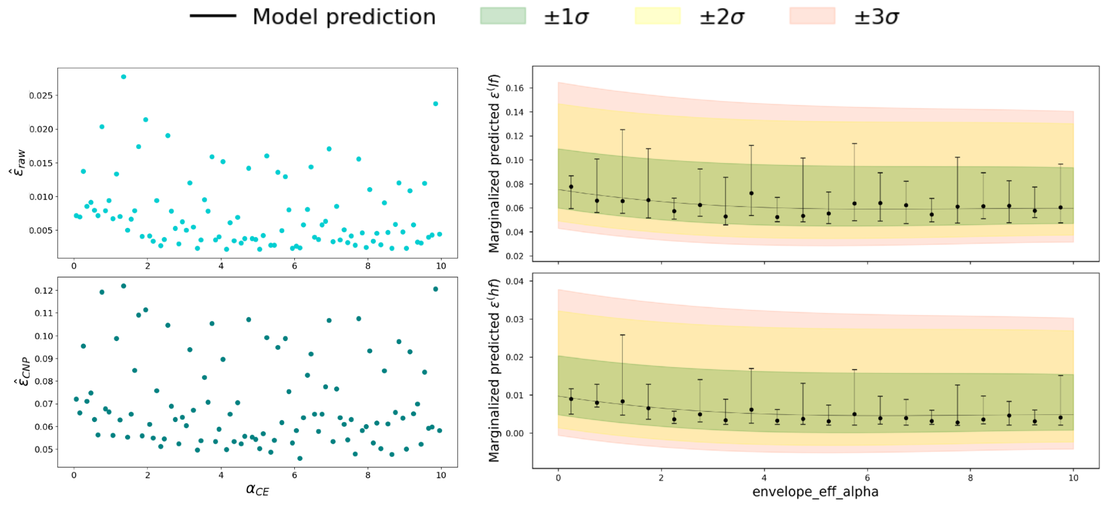


[Broekgaarden et al. (2022)]

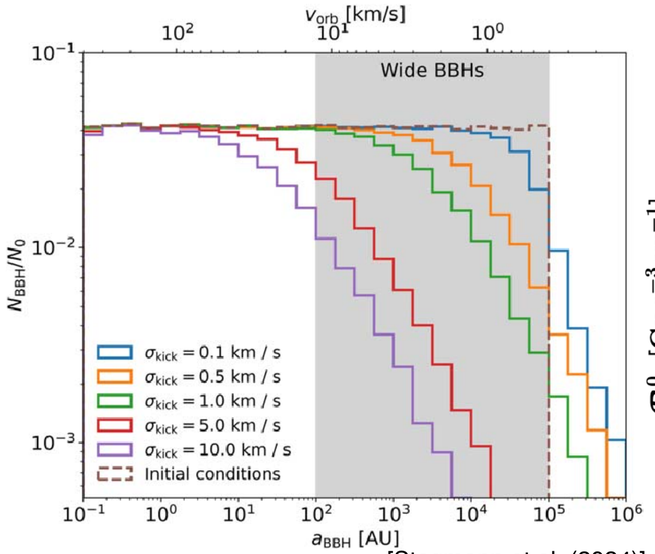
- Too low → binaries merge during CE phase → few DCOs
- Too high → post-CE orbits too wide → binaries don't merge in Hubble time
- Sweet spot $\alpha_{CE} \sim 0.5 - 2.0$ → most BBHs form and merge

Common Envelop determines whether the binary will survive and how tight its orbit will be after envelope ejection

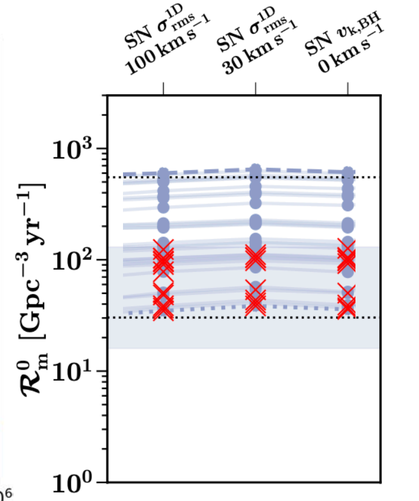
Model Result



What do we know about the dependance of the BBH merger rate on the natal kick?



[Stegmann et al. (2024)]



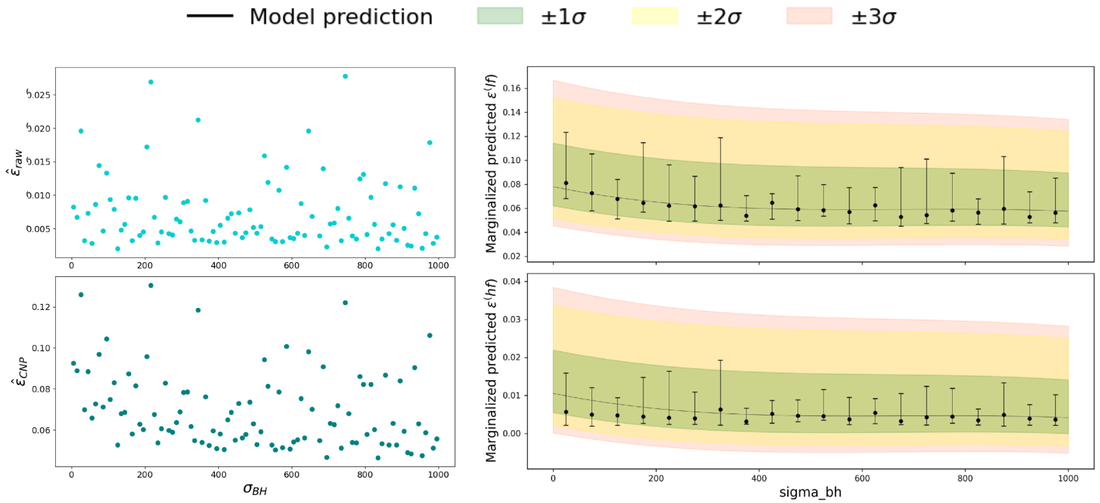
[Broekgaarden et al. (2022)]

- Higher natal kicks tend to suppress BBH merger rate
- Low kicks help binaries stay bound; high kicks disrupt systems
- Formation efficiency shows non-monotonotic dependence

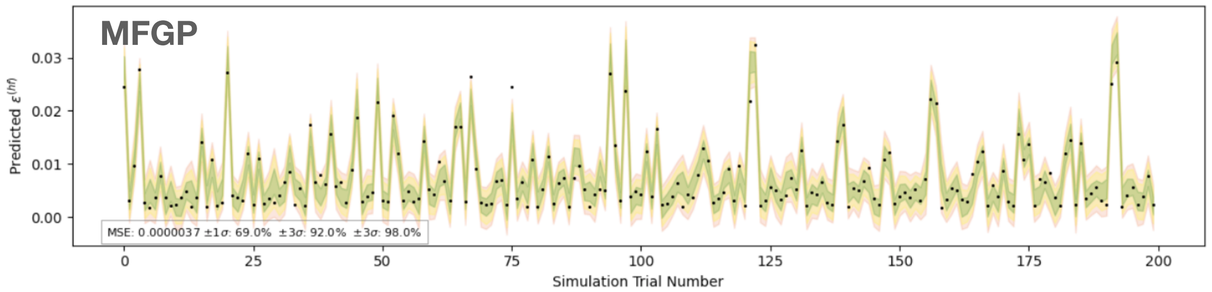
Compact objects are expected to receive a natal kick from the parent SN explosion (arise from asymmetries in SN ejecta and neutrino flux)

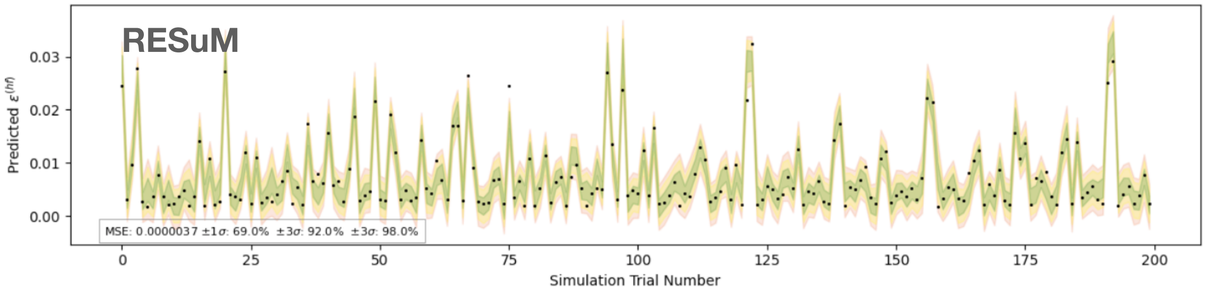
Kicks can **unbind binaries** or alter orbital properties (eccentricity, spin misalignment)

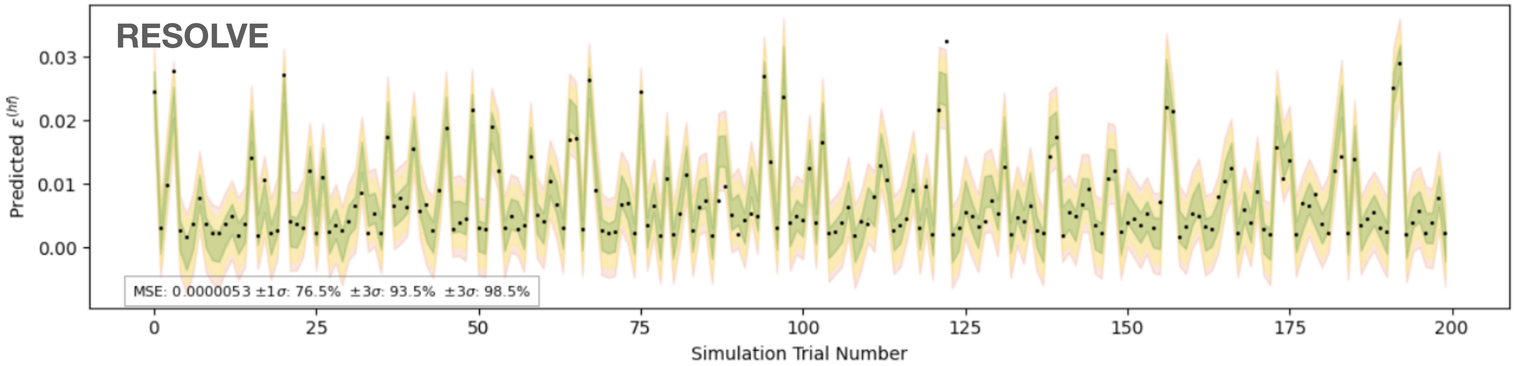
Model Result



Validation







Model	Dataset	(#LF,#HF)	MSE	$1\hat{\sigma}$ [%]	$2\hat{\sigma}$ [%]	$3\hat{\sigma}$ [%]
MFGP	Black Hole	(1000,5)	9.3×10^{-6}	72.0%	90.5%	96.5%
MFGP	Black Hole	(1000,10)	8.1×10^{-6}	68.5%	91.0%	96.0%
MFGP	Black Hole	(1000,15)	9.3×10^{-6}	64.5%	88.5%	95.0%
RESuM	Black Hole	(1000,5)	17.7×10^{-6}	49.5%	71.0%	78.5%
RESuM	Black Hole	(1000,10)	9.2×10^{-6}	51.0%	78.0%	85.5%
RESuM	Black Hole	(1000,15)	3.7×10^{-6}	69.0%	92.0%	98.0%
RESOLVE (d=1)	Black Hole	(1000,15)	15.3×10^{-6}	66.0%	94.5%	99.0%
RESOLVE (d=2)	Black Hole	(1000,15)	7.0×10^{-6}	72.0%	96.0%	98.5%
RESOLVE (d=4)	Black Hole	(1000,15)	5.0×10^{-6}	77.5%	96.0%	98.5%
RESOLVE (d=5)	Black Hole	(1000,15)	5.3×10^{-6}	76.0%	94.5%	98.5%
RESOLVE (d=3)	Black Hole	(1000,5)	18.6×10^{-6}	54.0%	75.0%	86.5%
RESOLVE (d=3)	Black Hole	(1000,10)	8.2×10^{-6}	70.5%	91.0%	97.0%
RESOLVE (d=3)	Black Hole	(1000,15)	5.3×10^{-6}	76.5%	93.5%	98.5%
Proper Statistical Coverage				68.27%	95.45%	99.73%

Proof of Concept!

Infer Posteriors from

Method:

- The BBH merger rate is computed by convoluting the formation efficiency with $\mathrm{SFRD(z)}$

We simplified using $z_{obs} = 0$ and $t_d = 0$, scaling with

$$SFRD(z = 0) \approx 0.015 \mathrm{M_{sol}} \mathrm{yr}^{-1} \mathrm{Mpc}^{-3}$$

- We constrain binary evolution parameters by matching predicted formation efficiency to the observed GW merger rate.

Results:

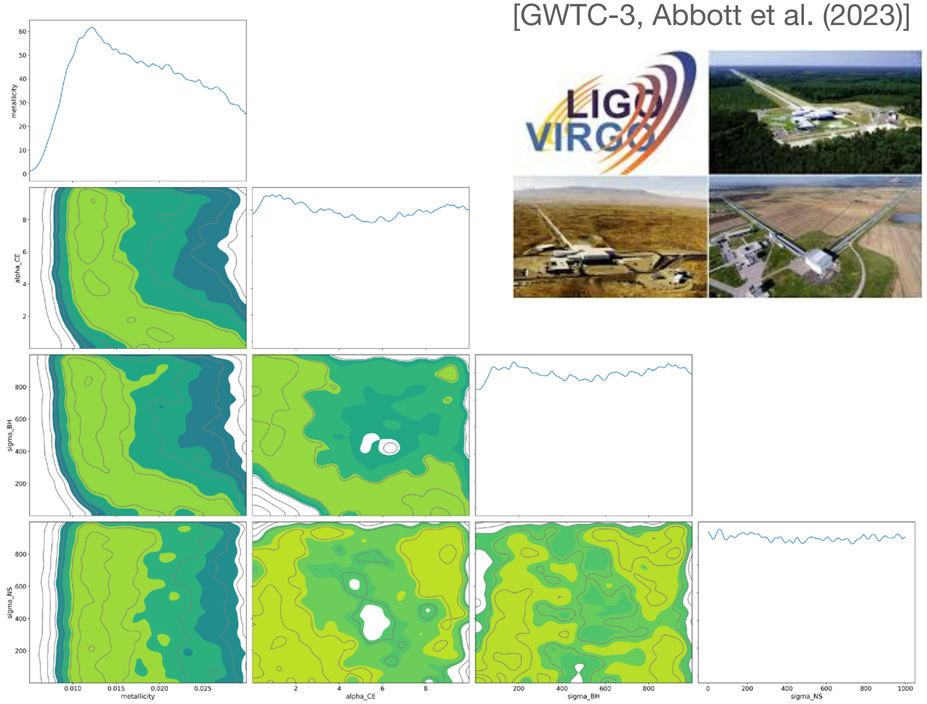
- Metallicity posterior peaks at $0.0172^{+0.0079}_{-0.0062}$ near solar $Z_{\odot}$, but with large uncertainty.
- α_{CE}, σ_{BH}, σ_{NS} remain weakly constrained due to known degeneracies.
- Results align with expectations from population synthesis studies (e.g., Neijssel+2019, Broekgaarden+2021).
- Simplified SFRD scaling adds further uncertainty, especially in Z

Compared to LIGO-Virgo's observed rate

LIGO-Virgo collaboration reported an observed gravitational wave rate of

$$y_{obs} = 17^{+10}_{-6.7} \mathrm{yr}^{-1} \mathrm{Gpc}^{-3}$$

[GWTC-3, Abbott et al. (2023)]



RESuM - Rare Event Surrogate Model for Detector Design

ICLR Spotlight

RESOLVE - Rare Event Surrogate Likelihood for Gravitational Wave Paleontology Parameter Estimation

arxiv:2507.15967

Thank you!

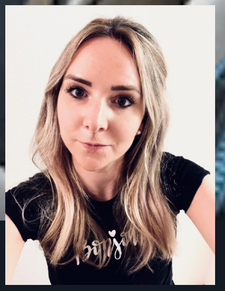
COMPAS

Alex Migala
UCSD

Adam Boesky
Harvard

Alan Poon
LBNL

Floor Broekgaarden
UCSD


Ann-Kathrin Schuetz
LBNL

Aobo Li
UCSD

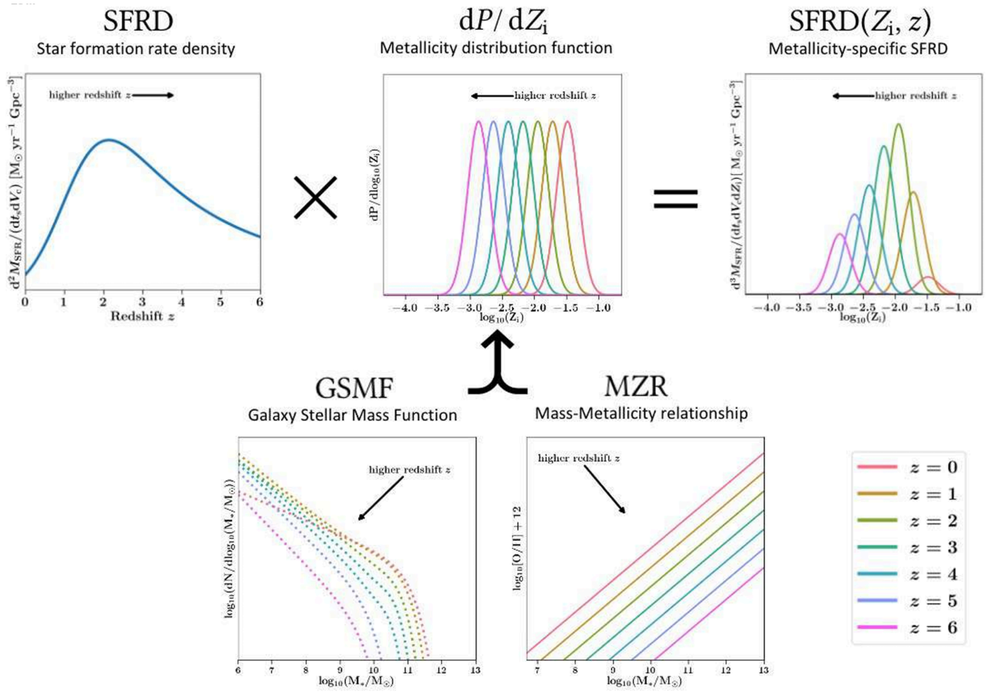
BBH Merger Rate

- The BBH merger rate is computed by convoluting the formation efficiency with$\mathrm{SFRD(z)}$ with **metallicity-dependent** dP/dZ_i
- Full model integrates over formation redshift with delay time:

$$R_{BBH}(z_{obs}) = \int_{z_{obs}}^{\infty} \mathrm{SFRD(z_{form})} \cdot \epsilon_{\mathrm{BBH}}(\mathrm{Z(z_{form})}) \cdot \mathrm{P(t(z_{form}) - t(z_{obs}))} \frac{\mathrm{dt}}{\mathrm{dz_{form}}} \mathrm{dz_{form}}$$

We simplified using $z_{obs} = 0$ and $t_d = 0$, scaling with

$$SFRD(z = 0) \approx 0.015 \mathrm{M_{sol} yr^{-1} Mpc^{-3}}$$



Standard fit from [Madau & Dickinson 2014]:

$$\mathrm{SFRD(z)} = 0.015 \frac{(1+\mathrm{z})^{2.7}}{1 + \left(\frac{1+\mathrm{z}}{2.9}\right)^{5.6}} \mathrm{M_{sol} yr^{-1} Mpc^{-3}}$$

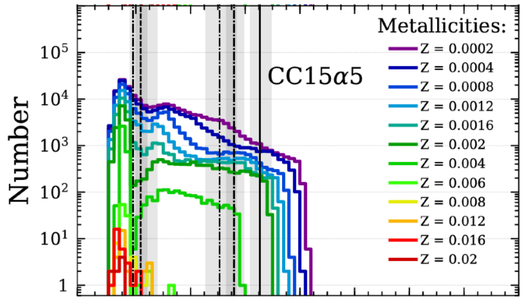
inclusion of metallicity-dependent SFRD → Future work

Parameters of interest (Hyperparameters)

Metallicity

- Metallicity strongly affects both **single and binary star evolution** through its impact on **stellar winds**

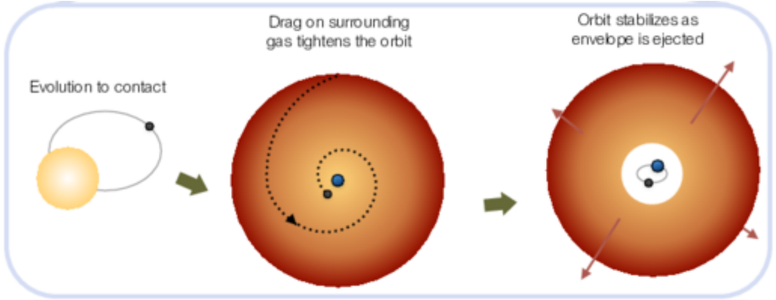
  (e.g. higher metallicity → stronger stellar winds → greater mass loss during stellar evolution)



[Giacobbo&Mapelli 2019]

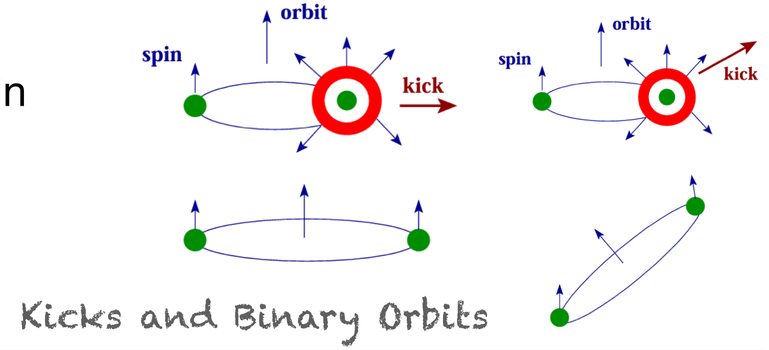
Common Envelope

- Common Envelop determines whether the binary will survive and how tight its orbit will be after envelope ejection

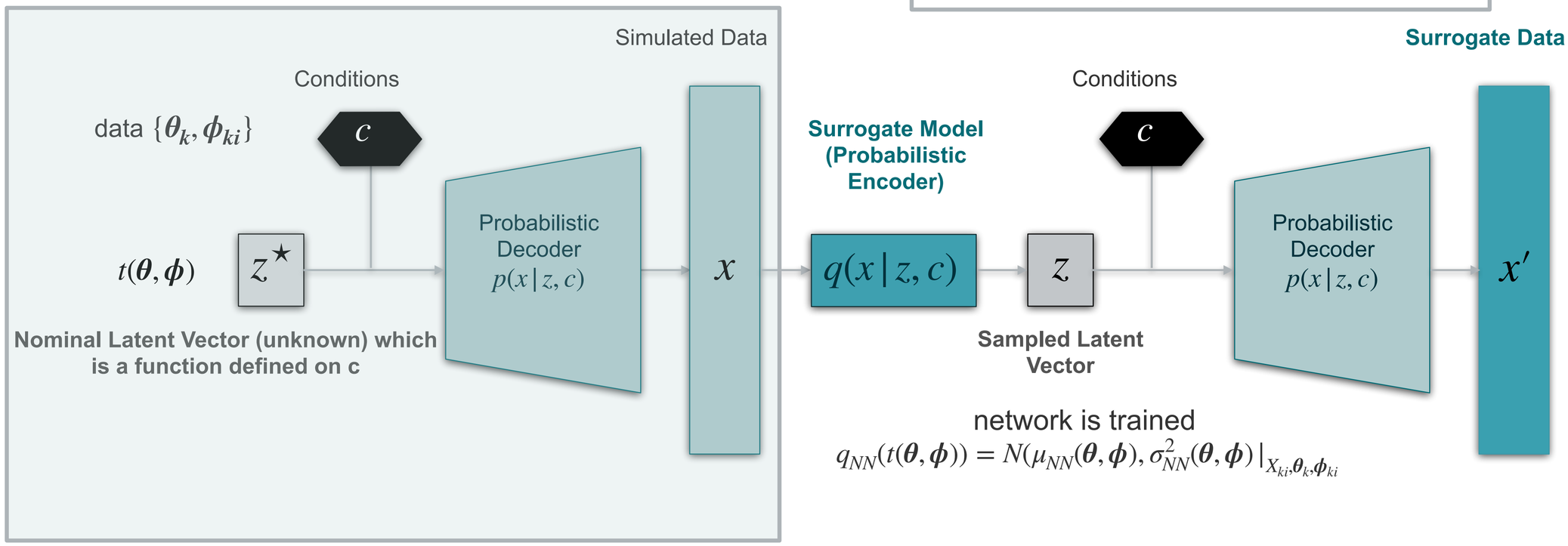


SN kick
BH kick

- Compact objects are expected to receive a natal kick from the parent SN explosion (arise from asymmetries in SN ejecta and neutrino flux)
- Kicks can **unbind binaries** or alter orbital properties (eccentricity, spin misalignment)



Posterior distribution $q(t(\theta, \phi))$ serves as probabilistic encoder



Bernoulli process acts as probabilistic decoder

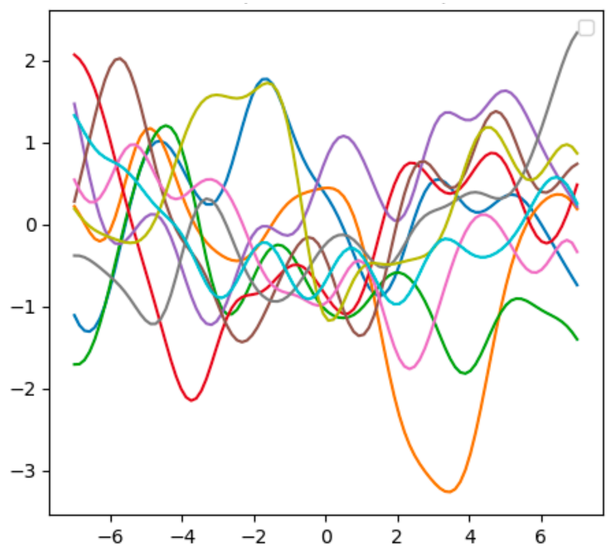
Bayes Theorem → minimizing the likelihood of the observed data $\mathscr{L}(X_{ki} \mid \theta_k, \phi_{ki}, q(t))$

→ regularization

Regression task:
predict the value of y_n for a new value of x_n where $f : \{x_n\}^N \rightarrow \{y_n\}^N$ maps the input space to the output space

Let's start with a distribution of all possible functions that, could have produced our data (without actually looking at the data!).

$$f(\cdot) \sim p(f(\cdot)) \sim \mathcal{N}(\mu(.), \sigma(.))$$



A Gaussian process is a probability distribution over possible functions that fit a set of points.

Gaussian process

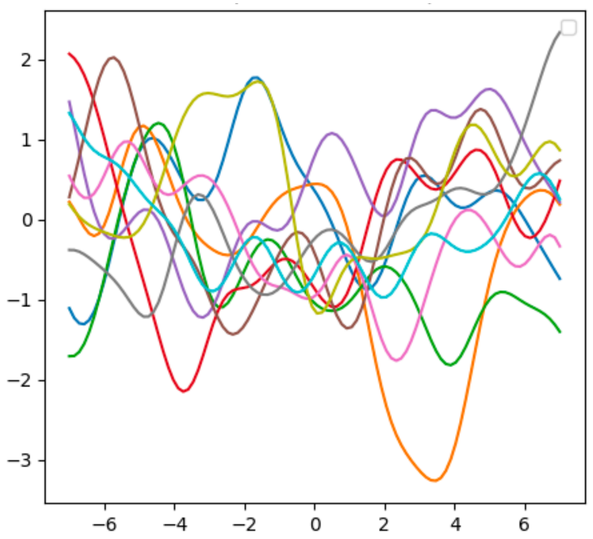
$$p(f(x) \mid I) = GP(f(x) \mid m(x), k(x, x'))$$

Anything we know (potentially observed input/ output data

Could be prior or posterior Mean/covariance

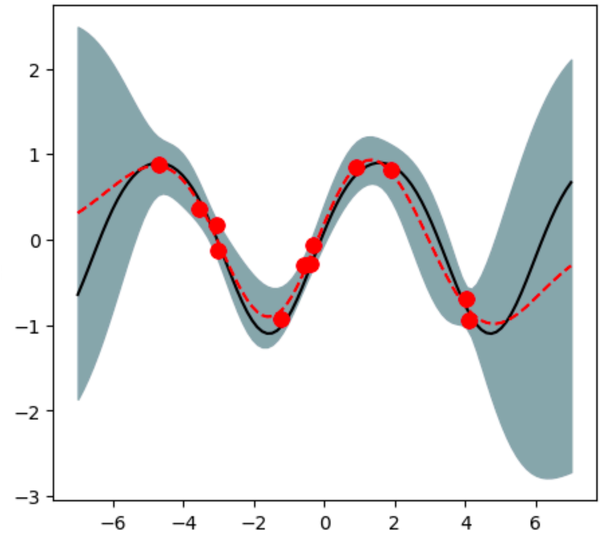
- pretty strong confidence regarding how the data-generating function should look like
- stating the model's uncertainty, being high where it lacks data and low where it has many data points

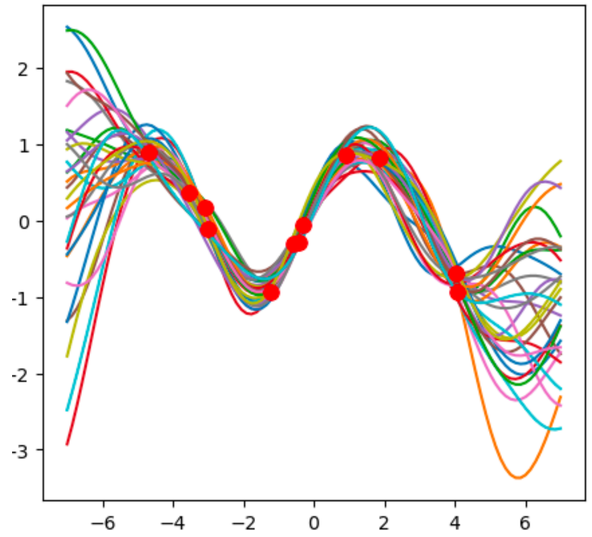
GP prior



Sampling from the GP prior after 10 noise free observations

GP posterior





Regression task:

predict the value of y_n for a new value of x_n where $f : \{\theta_n\}^N \rightarrow \{y_n\}^N$ maps the input space to the output space

Let's start with a distribution of all possible functions that, could have produced our data (without actually looking at the data!).

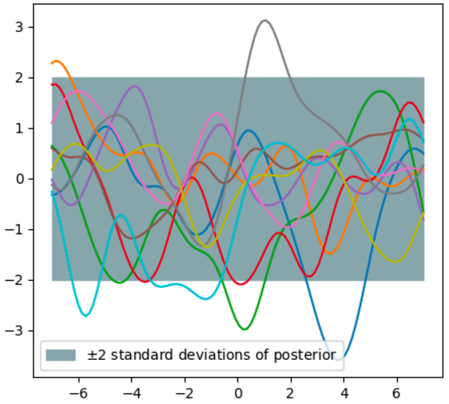
$$f(\cdot) \sim p(f(\cdot)) \sim \mathcal{N}(\mu(.), \sigma(.))$$

A Gaussian process is a probability distribution over possible functions that fit a set of points.

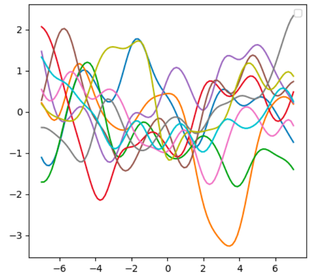
Gaussian Process

$$f(\theta) \sim GP(m(\theta), k(\theta, \theta'))$$

mean of posterior



GP prior



data

GP posterior